Exercise Solutions

Question 1

Suppose we have a simplified language with:

- Three vowels: a, i, and u
- Three consonants: p, t, and k

Let X map consonants $\{p, t, k\}$ to range $R_X = \{1, 2, 3\}$
Let Y map vowels $\{a, i, u\}$ to range $R_Y = \{1, 2, 3\}$

Suppose we estimated the joint PMF for vowel and consonant co-occurrences within syllables, as given below.

$$P(x,y) = \begin{array}{c|c|c|c} y \setminus x & 1 & 2 & 3 \\ \hline 1 & \frac{1}{16} & \frac{3}{8} & \frac{1}{16} \\ 2 & \frac{1}{16} & \frac{3}{16} & 0 \\ 3 & 0 & \frac{3}{16} & \frac{1}{16} \end{array}$$

Questions: Calculate the following:

- Entropies $H(X)$ and $H(Y)$
- Conditional entropies $H(X|Y)$ and $H(Y|X)$
- Joint entropy $H(X, Y)$
- Mutual information $I(X; Y)$

Answer for Question 1

It will be useful to calculate the marginal probabilities from the given joint PMF table:

$$P(x,y) = \begin{array}{c|c|c|c|c} y \setminus x & 1 & 2 & 3 & P(y) \\ \hline 1 & \frac{1}{16} & \frac{3}{8} & \frac{1}{16} & \frac{1}{2} \\ 2 & \frac{1}{16} & \frac{3}{16} & 0 & \frac{1}{4} \\ 3 & 0 & \frac{3}{16} & \frac{1}{16} & \frac{1}{4} \\ \hline P(x) & \frac{1}{8} & \frac{3}{4} & \frac{1}{8} & 1 \end{array}$$

- $H(X) = -\sum_i P(x_i) \log_2 P(x_i)$

$$H(X) = -\left(\frac{1}{8}\log_2\frac{1}{8} + \frac{3}{4}\log_2\frac{3}{4} + \frac{1}{8}\log_2\frac{1}{8}\right) = 1.061 \text{ bits}$$

- $H(Y) = -\sum_j P(y_j) \log_2 P(y_j)$

$$H(Y) = -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{4}\log_2\frac{1}{4}\right) = 1.5 \text{ bits}$$

- Joint Entropy:

$$H(X,Y) = -\sum_{x\in R_X}\sum_{y\in R_Y} P(x,y)\log P(x,y) = 2.436 \text{ bits}$$

- Conditional Entropies:

$$H(X|Y) = H(X,Y) - H(Y) = 2.436 - 1.5 = 0.936 \text{ bits}$$

$$H(Y|X) = H(X,Y) - H(X) = 2.436 - 1.061 = 1.375 \text{ bits}$$

- Mutual Information:

$$I(X;Y) = H(X) - H(X|Y) = 1.061 - 0.936 = 0.125 \text{ bits}$$

Question 2: From the 2021 AI2 exam paper

As a data scientist in a telecommunication company, your task is to analyze a customer dataset to predict whether a customer will terminate his/her contract. The dataset consists of around 8000 customer records, each consisting of one binary dependent variable Y, indicating whether the customer terminates the contract ($Y = 1$) or not ($Y = 0$), and 19 independent variables, which include the customer's information, e.g., age, subscription plan, extra data plan, etc., and consumer behavior such as the average number of calls and hours per week.

Since your boss needs some actionable insights to retain customers, you decided to use interpretable machine learning methods. Design your interpretable machine learning method by answering the following questions:

(a)

You have implemented a feature selection algorithm based on mutual information to select the most informative features from the 19 independent variables. To validate the implementation of your mutual information calculation function, you use a small subset of the data to calculate mutual information manually. You select one independent variable *subscription plan*, denoted as S, which takes two values, $S \in \{1, 2\}$. Please use the following Probability Mass Function table:

$p(S,Y)$	$S=1$	$S=2$
$Y=0$	$\frac{2}{12}$	$\frac{5}{12}$
$Y=1$	$\frac{2}{12}$	$\frac{3}{12}$

Calculate:

- Entropies $H(S)$ and $H(Y)$
- Conditional entropies $H(S|Y)$ and $H(Y|S)$
- Joint entropy $H(S,Y)$
- Mutual information $I(S;Y)$

Show all your working. Discuss what mutual information means and whether this feature will be selected or not.

(b)

After applying your algorithm, you selected two variables:

- Extra data plan E, which is a binary random variable that indicates whether the customer subscribes to the extra data plan ($E=1$) or not ($E=0$).
- Averaged hours used per week H, which is a continuous random variable.

You then built a logistic regression model to classify customers into *low risk* or *high risk* of terminating the contract. The fitted model is:

$$\log\left(\frac{p}{1-p}\right) = -0.77 + 0.23H - 1.18E$$

i) Given a customer $\mathbf{x}$ who has the extra data plan ($E=1$) and spent on average 0.5 hours per week, calculate the odds and the probability that the customer will terminate the contract ($Y=1|\mathbf{x}$).
ii) Using this fitted model, explain to your boss what actions should be taken to retain customers.

Answers for Question 2

(a)

Given the probability mass function:

$p(S,Y)$	$S=1$	$S=2$	$p(Y)$
$Y=0$	$\frac{2}{12}$	$\frac{5}{12}$	$\frac{7}{12}$
$Y=1$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{5}{12}$
$p(S)$	$\frac{4}{12}$	$\frac{8}{12}$	1

Calculating entropies:

$$H(S) = -\sum p(s_i)\log_2 p(s_i) = -\left(\frac{4}{12}\log_2\frac{4}{12} + \frac{8}{12}\log_2\frac{8}{12}\right) = 0.92 \text{ bits} \quad (1)$$

$$H(Y) = -\sum p(y_i)\log_2 p(y_i) = -\left(\frac{7}{12}\log_2\frac{7}{12} + \frac{5}{12}\log_2\frac{5}{12}\right) = 0.97 \text{ bits} \quad (2)$$

$$H(S,Y) = -\sum_{s_i\in R_S, y_j\in R_Y} p(s_i,y_j)\log p(s_i,y_j) = 1.89 \text{ bits} \quad (3)$$

$$H(S|Y) = H(S,Y) - H(Y) = 1.89 - 0.97 = 0.92 \text{ bits} \quad (4)$$

$$H(Y|S) = H(S,Y) - H(S) = 1.89 - 0.92 = 0.97 \text{ bits} \quad (5)$$

$$I(S;Y) = H(S) - H(S|Y) = 0.92 - 0.92 = 0 \text{ bits} \quad (6)$$

Since mutual information measures the shared information between S and Y, a value of 0 means that knowing S does not reduce uncertainty about Y. Therefore, this feature is not useful for prediction.

(b)

The odds:

$$\begin{aligned}\text{odds} &= \exp(-0.77 + 0.23\times 0.5 - 1.18)\\ &= \exp(-0.77 + 0.115 - 1.18) = 0.1596\end{aligned}$$

The probability (note that $p = \frac{1}{1+\frac{1}{odds}}$):

$$P(Y=1|\mathbf{x}) = \frac{1}{1+\frac{1}{0.1596}} = 0.137$$

We analyze the impact of independent variables:

$$OR_E = \frac{\text{odds when } E=1}{\text{odds when } E=0} = \frac{\exp(-0.77+0.23H-1.18)}{\exp(-0.77+0.23H)} \quad (7)$$

$$= \exp(-1.18) \approx 0.31 \quad (8)$$

$$
\begin{aligned}
OR_H &= \frac{\text{odds when } H = h + \Delta}{\text{odds when } H = h} && (9)\\
&= \frac{\exp(-0.77 + 0.23(h + \Delta) - 1.18E)}{\exp(-0.77 + 0.23h - 1.18E)} && (10)\\
&= \exp(0.23\Delta) && (11)\\
&\approx 1.25^{\Delta} && (12)
\end{aligned}
$$

From the analysis, we can suggest to the boss that:
- If a customer adds the extra data plan, the odds of terminating the contract decrease by a factor of 3.
- If a customer increases the average time by one hour, the odds of terminating the contract increase by a factor of 1.25.

So, from the analysis, we can suggest to the boss that, the more hours the customers spent, the more likely the customers will terminate, which means the company should improve its telecommunication service/price. However, by simply persuading them to subscribe to the extra data plan, they are more likely to stay.

This implies that improving the telecommunication service/price for high-usage customers is essential to reduce churn, while encouraging extra data plan subscriptions may enhance customer retention.

Question 3: From the 2022 AI2 exam paper

As a machine learning expert for an AI cybersecurity company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\mathbf{X} = [X_1, X_2, \ldots, X_n]$. Each record also includes a binary random variable Y called label that was labeled by cybersecurity experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

Answer the following questions about Feature Selection Based on Mutual Information.

(i) Explain to your colleague, who knows nothing about information theory, the concept of mutual information.

(ii) Explain the loop in the pseudocode of Table 1.

Answers to Question 3

(i) Mutual information measures the amount of information that one random variable provides about another. It quantifies how much knowing one variable reduces uncertainty about the other. High mutual information indicates strong

1.	Initialize: Set $F \leftarrow \mathbf{X}$ and $S \leftarrow \emptyset$
2.	Select $f_{max} = \arg\max_{X_i \in X} I(Y; X_i)$
3.	Update $F \leftarrow F \setminus \{f_{max}\}$, $S \leftarrow \{f_{max}\}$
4.	Repeat until $\lvert S \rvert = K$:
5.	Select $f_{max} = \arg\max_{X_i \in F} \left[I(Y; X_i) - \sum_{X_s \in S} I(X_s; X_i) \right]$
6.	Update $F \leftarrow F \setminus \{f_{max}\}$, $S \leftarrow S \cup \{f_{max}\}$

Table 1: Pseudocode of Mutual Information based Feature Selection Algorithm

dependence between variables, while low mutual information suggests independence.

(ii) The two lines in the loop are used to select K features. In this loop, we find the feature $f_{\max}$ which achieves the maximum mutual information I among all the remaining independent variables in set F. However, some features may be highly correlated with each other, hence selecting them will increase the number of features without improving the prediction. Therefore, we make sure that there is minimal redundancy between the candidate feature X_i and the set of selected features S. That's exactly what the second term on the RHS achieves in line 5. We then add this feature into the set S and subtract it from set F, and repeat until we got K features.

In other words, the loop iterates to select the best K features by maximizing mutual information while minimizing redundancy.

(The whole algorithm does the following:

-Initially selecting the feature with the highest mutual information with Y.

-Iteratively selects the feature that maximizes mutual information with Y while ensuring minimal redundancy with already selected features.

-Repeats until K features are chosen.)

Question 4: Decision Tree Classification

A company wants to classify whether a customer will purchase a product (Y = Yes) or not (Y = No) based on categorical features. You have access to the following dataset:

Age Group	Income Level	Prior Purchase	Purchase (Y)
Young	Low	No	No
Young	Low	Yes	No
Young	High	No	Yes
Middle-aged	Low	No	No
Middle-aged	High	No	Yes
Senior	Low	No	No
Senior	High	Yes	Yes
Senior	High	No	Yes

(a) Calculate the entropy of the target variable Y.

(b) Compute the information gain for splitting on the feature "Prior Purchase."

(c) If you were to build a decision tree, which feature would be the best root node? Justify your answer.

(d) Draw the first split of the decision tree based on the best attribute.

(e) Discuss how overfitting can be avoided in decision trees and suggest techniques to improve generalization.

Answer to Question 4

(a) Entropy of Y:

$$H(Y) = -p_+ \log_2(p_+) - p_- \log_2(p_-) \tag{13}$$

where $p_+ = 4/8$ and $p_- = 4/8$, leading to:

$$H(Y) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1.0 \tag{14}$$

(b) Information gain for "Prior Purchase": Splitting results in two subsets:

- For Prior Purchase = Yes: [No, Yes]
- For Prior Purchase = No: [No, No, Yes, No, Yes, Yes]

We see that "Prior Purchase" makes no reduction in entropy.

IG(Prior Purchase)=0. Not a good feature to split on!

(c) The best root node should have the highest information gain. "Income Level" has maximal IG, so it should be selected as the root node.

(d) Decision tree first split:

- If Income Level = Low $\Rightarrow$ Y = No
- If Prior Purchase = High $\Rightarrow$ Y=Yes.

(e) Techniques to avoid overfitting:

- Pruning: Remove branches that add little classification value. Use a validation set to find how much to prune.
- Limiting depth: Restrict maximum depth of the tree. Use a validation set to find when to stop growing the tree.
- Applying ensemble methods: Use bagging or boosting.